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Practical No 1 Date :- June/20/2023

Find confidence interval of mean assuming normal distribution for following data.

Height: 78 55 68 48 65 76 57 55 65 75 51 61 68 67 76 78 71 56 57 67 58 51 50 58 50 77 55 48 70 55 58 70 56 52 74 61 69 76 61 68 78 56 78 57 66 66 74 66 48 73 71 70 62 74 76 50 69 75 65 48

➔ Solution :-

Descriptive

			Statistic	Std. Error
Height	Mean		63.88	1.233
	95% Confidence Interval for Mean	Lower Bound	61.42	
		Upper Bound	66.35	
	5% Trimmed Mean		63.98	
	Median		65.50	
	Variance		91.257	
	Std. Deviation		9.553	
	Minimum		48	
	Maximum		78	
	Range		30	
	Interquartile Range		17	
	Skewness		-.136	.309
	Kurtosis		-1.258	.608

➔Hence, the confidence interval of mean for the given data is (61.42, 66.35).

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Practical No 2 Date :-June/20/2023

The time (in minutes) spent by 10 randomly selected customers using internet in a cyber cafe is as follows: 35,20,30,45,60,40,65,40,25,50. Can you say that the average time spent by customers is more than 30 minutes at 5% level of significance?

➔Solution:

- ⇨ Null Hypothesis: H0: μ = 30 (The average time spent by customers is 30 minutes at 5% level of significance.)
- ⇨ Alternate Hypothesis: H1: μ>30(The average time spent by customers is more than 30 minutes at 5% level of significance.)

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
timeinminutes	10	41.00	14.491	4.583

One-Sample Test

	Test Value = 30					
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
timeinminutes	2.400	9	.040	11.000	.63	21.37

α= 5% i.e. α=0.05

2p=0.040

p=0.020

➔Since, p (0.020) < α (0.05) we accept H1 at 5% level of significance so we are 95% sure that the average time spent by customers is more than 30 minutes.

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Practical No 3 Date:- June/21/2023

Two kinds of manure were applied to sixteen one-hectare plot, other condition remaining the same. The yields in quintals are given below:

Manure I	18	20	36	50	49	36	34	49	41
Manure II	29	28	26	35	30	44	46		

Is there any significant difference between the mean yields? Use 5% level of significance.

➔Solution:

- ⇨ Null Hypothesis: H0: μ1= μ2 (There is no significant difference between the mean yields at 5% level of significance.)
- ⇨ Alternate Hypothesis: H1: μ1≠ μ2 (There is significant difference between the mean yields at 5% level of significance.)

Group Statistics

	type	N	Mean	Std. Deviation	Std. Error Mean
value	Manure I	9	37.00	11.906	3.969
	Manure II	7	34.00	8.021	3.032

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
									95% Confidence Interval of the Difference	
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
value	Equal variances assumed	.756	.399	.571	14	.577	3.000	5.251	-8.262	14.262
	Equal variances not assumed			.601	13.797	.558	3.000	4.994	-7.726	13.726

α = 5% i.e. α = 0.05

p = 0.577

➔Since, p (0.577) > α (0.05) so we accept H0 at 5% level of significance.

Therefore, we are 95% sure that there is no significant difference between the mean yields of manure I and manure II.

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Practical No 4 Date:- June/21/2023

Memory capacity of 10 students was tested before and after training, state whether the training was effective or not from the following scores:

Roll No.	1	2	3	4	5	6	7	8	9	10
Before training	12	14	11	8	7	10	3	0	5	6
After training	15	16	10	7	5	12	10	2	3	8

➔Solution:

- ⇨ Null Hypothesis: H0: μd=0(The training was not effective at 5% level of significance.)
- ⇨ Alternate Hypothesis: H1: μd<0(The training was effective at 5% level of significance.)

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	BeforeTraining	7.60	10	4.300	1.360
	AfterTrainnig	8.80	10	4.733	1.497

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	BeforeTraining & AfterTrainnig	10	.815	.004

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	BeforeTraining - AfterTrainnig	-1.200	2.781	.879	-3.189	.789	-1.365	9	.206

α= 5% i.e. 0.05

2p=0.206

P=0.103

➔Since, p (0.103) > α (0.05) so we accept H0 at 5% level of significance. Therefore, we are 95% sure that the training was not effective.

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Practical No 5 Date:- June/22/2023

In 30 toss of a coin the following sequence of heads (H) and tails (T) is obtained.
H T T H T H H H T H H T T H T H T H H T H T T H T H H T H T
Test at 0.05 level of significance level whether the sequence is random.

➔Solution:

- ⇨ Null Hypothesis: H0: The sequence is random.
- ⇨ Alternate Hypothesis: H1: The sequence is not random.

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
outcome	30	.53	.507	0	1	.00	1.00	1.00

Runs Test

	outcome
Test Value[a]	1
Cases < Test Value	14
Cases >= Test Value	16
Total Cases	30
Number of Runs	22
Z	2.078
Asymp. Sig. (2-tailed)	.038

a. Median

α = 5% i.e. α = 0.05
p = 0.038
➔Since, p (0.038) < α (0.05) so we accept H1 at 5% level of significance.
Therefore, we are 95% sure that the sequence is not random.

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Practical No 6 Date:- June/22/2023

Test whether the coin is unbiased from the following observations.

Tail	Tail	Head	Head	Tail	Head	Tail	Head	Head	Tail
Head	Head	Head	Tail	Head	Head	Head	Head	Head	Head
Tail	Tail	Tail	Tail	Head	Tail	Tail	Tail	Tail	Tail
Tail	Tail	Head	Tail	Tail	Tail	Head	Tail	Tail	Tail
Tail	Head	Tail	Tail	Head	Tail	Head	Tail	Tail	Tail

➔Solution:

- ⇨ Null Hypothesis: H0: P=0.5 (The coin is unbiased.)
- ⇨ Alternate Hypothesis: H1:P≠0.5 (The coin is not unbiased.)

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
outcome	50	.40	.495	0	1	.00	.00	1.00

Binomial Test

		Category	N	Observed Prop.	Test Prop.	Exact Sig. (2-tailed)
outcome	Group 1	T	30	.60	.50	.203
	Group 2	H	20	.40		
	Total		50	1.00		

α = 5% i.e. α = 0.05

p = 0.203

➔Since, p (0.203) > α (0.05) so we accept H0 at 5% level of significance.

Therefore, we are 95% sure that the sequence is unbiased.

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Practical No 7 Date:- June/23/2023

The number of disease infected tomato plants in 10 different plots of equal size are given below. Test whether the disease plants are uniformly distributed over the entire area use Kolmogorov Smirnov test.

Plot No.	1	2	3	4	5	6	7	8	9	10
No of infected Plants	8	10	9	12	15	7	5	12	13	9

➔Solution:

- ⇨ Null Hypothesis: H0: The disease plants are uniformly distributed over the entire area.
- ⇨ Alternate Hypothesis: H1: The disease plants are not uniformly distributed over the entire area.

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
NoofInfrctedPlants	10	10.00	3.018	5	15	7.75	9.50	12.25

One-Sample Kolmogorov-Smirnov Test

		NoofInfrctedPlants
N		10
Normal Parameters[a,b]	Mean	10.00
	Std. Deviation	3.018
Most Extreme Differences	Absolute	.146
	Positive	.130
	Negative	-.146
Kolmogorov-Smirnov Z		.462
Asymp. Sig. (2-tailed)		.983

a. Test distribution is Normal.

b. Calculated from data.

α= 5% i.e. α=0.05 , p=0.963

➔Since, p (0.983) > α (0.05) so we accept H0 at 5% level of significance.

Therefore, we are 95% sure that the disease infected plants are uniformly distributed over the entire area.

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Practical No 8 Date :- June/23/2023

Test whether the given data of weight is normally distributed or not.

19.5	20	26.9	27.1	28.1	30	31.6	32.7	34.4	37.2
37.5	37.9	38	38.4	38.6	38.8	38.9	40.1	41.6	42.6
42.9	45	45.2	45.5	46.5	46.8	47.3	48.1	48.3	48.4
48.8	49	49.1	49.3	49.4	49.5	49.9	50.4	51.8	54.4
54.9	55.3	55.6	57.3	57.4	57.5	58.7	58.8	59.7	59.9

➔Solution:

- ⇨ Null Hypothesis: H0: The given weight is normally distributed.
- ⇨ Alternate Hypothesis: H1: The given weights is not normally distributed.

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
weight	50	44.412	10.3139	19.5	59.9	37.975	46.650	50.750

One-Sample Kolmogorov-Smirnov Test

		weight
N		50
Normal Parameters[a,b]	Mean	44.412
	Std. Deviation	10.3139
Most Extreme Differences	Absolute	.103
	Positive	.067
	Negative	-.103
Kolmogorov-Smirnov Z		.726
Asymp. Sig. (2-tailed)		.667

a. Test distribution is Normal.

b. Calculated from data.

α = 5% i.e. α = 0.05

p = 0.667

➔Since, p (0.667) > α (0.05) so we accept H0 at 5% level of significance.

Therefore, we are 95% sure that the given weight is normally distributed.

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Practical No 9 Date:- June/23/2023

Test the hypothesis of no difference between the ages of male and female employees of a certain company, using the Mann-Whitney U test for the sample data below. Use α = 0.1.

Male	35	43	26	44	40	42	33	38	25	26
Female	30	41	34	31	36	32	25	47	28	24

➔Solution:

- ⇨ Null hypothesis: H0: There is no difference between the ages of male and female employees.
- ⇨ Alternate hypothesis: H1: There is difference between the ages of male and female employees.

Ranks

	Gender	N	Mean Rank	Sum of Ranks
Age	Male	10	11.65	116.50
	Female	10	9.35	93.50
	Total	20		

Test Statistics[a]

	Age
Mann-Whitney U	38.500
Wilcoxon W	93.500
Z	-.870
Asymp. Sig. (2-tailed)	.384
Exact Sig. [2*(1-tailed Sig.)]	.393[b]

a. Grouping Variable: Gender

b. Not corrected for ties.

α = 0.1

p = 0.384

➔Since, p (0.384) > α (0.1) we accept H0 at 10% level of significance.

Therefore, we are 90% sure that there is no difference between the ages of male and female employees.

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Practical No 10 Date:- July/13/2023

Data below shows one-week growth (in cm) of maize plant from two different localities:

Sample I	10	11	8	8	14		
Sample II	9	12	13	9	15	9	17

Test whether the two samples have come from the same population with respect to their medians. Use median test at 0.05 level of significance.

➔Solution:

- ⇨ Null hypothesis: H0: Md1 = Md2 (The two samples have come from the same population with respect to their medians.)
- ⇨ Alternate hypothesis: H1: Md1 ≠ Md2 (The two samples have come from the same population with respect to their medians.)

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
value	12	11.25	2.958	8	17	9.00	10.50	13.75
sample	12	1.58	.515	1	2	1.00	2.00	2.00

Frequencies

		sample	
		Sample I	Sample II
value	> Median	2	4
	<= Median	3	3

Test Statistics[a]

	value
N	12
Median	10.50
Exact Sig.	1.000

a. Grouping Variable: sample

α = 0.05

p = 1

➔Since, p (1) > α (0.05) we accept H0 at 5% level of significance.

Therefore, we are 95% sure the two samples have come from the same population with respect to their medians.

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Practical No 11 Date:- July/13/2023

Use Wilcoxon Matched pair signed rank test to determine the equality of effectiveness of two types of drugs in suppressing pain from following data.

Patient No.	Drug A	Drug B	Patient No.	Drug A	Drug B
1	6.5	3.5	11	5.4	5.5
2	3.7	3.7	12	4.0	4.1
3	3.9	4.7	13	5.7	4.1
4	6.7	5.0	14	3.9	4.2
5	6.2	5.6	15	3.6	3.7
6	6.7	4.3	16	4.9	4.1
7	6.1	5.4	17	3.9	5.4
8	4.3	5.8	18	5.8	3.7
9	5.5	4.3	19	4.9	4.1
10	6.8	4.3	20	4.9	4.1

➔Solution:

- ⇨ Null hypothesis: H0: Md1 = Md2 (The effectiveness of two types of drugs in suppressing pain is equal.)
- ⇨ Alternate hypothesis: H1: Md1 ≠ Md2 (The effectiveness of two types of drugs in suppressing pain is not equal.)

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
DrugA	20	5.170	1.1164	3.6	6.8	3.925	5.150	6.175
DrugB	20	4.480	.7157	3.5	5.8	4.100	4.250	5.300

Ranks

		N	Mean Rank	Sum of Ranks
DrugB - DrugA	Negative Ranks	12[a]	12.21	146.50
	Positive Ranks	7[b]	6.21	43.50
	Ties	1[c]		
	Total	20		

a. DrugB < DrugA

b. DrugB > DrugA

c. DrugB = DrugA

Test Statistics[a]

	DrugB - DrugA
Z	-2.076[b]
Asymp. Sig. (2-tailed)	.038

a. Wilcoxon Signed Ranks Test

b. Based on positive ranks.

α = 0.05

p = 0.038

➔Since, p (0.038) < α (0.05) we accept H1 at 5% level of significance.

Therefore, the effectiveness of two types of drugs in suppressing pain is not equal.

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Practical No 12 Date:- July/14/2023

Five housewives were asked for the acceptability of four brands of lipsticks for daily use. The response of acceptability (A) and rejection (R) are given below:

Housewives	Lipstick Brands			
	Alfa	Beta	Gamma	Delta
H1	A	R	A	R
H2	R	A	A	R
H3	R	A	R	A
H4	A	R	R	R
H5	A	A	R	A

Test whether there is any significant difference between brands with respect to acceptability.

➔Solution:

- ⇨ Null hypothesis: H0: There is no significant difference between brands with respect to acceptability.
- ⇨ Alternate hypothesis: H1: There is significant difference between brands with respect to acceptability.

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
Alfa	5	.60	.548	0	1	.00	1.00	1.00
Beta	5	.60	.548	0	1	.00	1.00	1.00
Gamma	5	.40	.548	0	1	.00	.00	1.00
Delta	5	.40	.548	0	1	.00	.00	1.00

Frequencies

	Value	
	0	1
Alfa	2	3
Beta	2	3
Gamma	3	2
Delta	3	2

Test Statistics

N	5
Cochran's Q	.667[a]
df	3
Asymp. Sig.	.881

a. 1 is treated as a success.

α = 0.05

p = 0.881

➔Since, p (0.881) > α (0.05) we accept H0 at 5% level of significance.

Therefore, there is no significant difference between brands with respect to acceptability.

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Practical No 13 Date:- July/14/2023

A survey was conducted in four hospitals in a particular city to obtain the number of babies born over a 12 months' period. This time period was divided into four seasons to test the hypothesis that the birth rate is constant over all the four seasons. The results of the survey were as follows:

Hospital	No of births			
	Winter	Spring	Summer	Fall
A	92	72	94	77
B	15	16	10	17
C	58	71	51	62
D	19	26	20	18

Analyze the data using Friedman two-way ANOVA test.

➔Solution:

- ⇨ Null hypothesis: H0: Md1 = Md2 = Md3 = Md4 (The birth rate is constant over all the four seasons.)
- ⇨ Alternate hypothesis: H1: At least one Mdi is different, i=1, 2, 3, 4 (The birth rate is constant over all the four seasons.)

Descriptive Statistics

	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
Winter	4	46.00	36.286	15	92	16.00	38.50	83.50
Spring	4	46.25	29.443	16	72	18.50	48.50	71.75
Summer	4	43.75	37.775	10	94	12.50	35.50	83.25
Fall	4	43.50	30.643	17	77	17.25	40.00	73.25

Ranks

	Mean Rank
Winter	2.25
Spring	3.00
Summer	2.25
Fall	2.50

Test Statistics[a]

N	4
Chi-Square	.900
df	3
Asymp. Sig.	.825

a. Friedman Test

α = 0.05

p = 0.825

➔Since, p (0.825) > α (0.05) we accept H0 at 5% level of significance.

Therefore, we are 95% sure that the birth rate is constant over all the four seasons.

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Practical No 14 Date:- July/21/2023

The table shows the corresponding values of the three variables X1, X2 and X3.

X1	X2	X3
5	12	51
7	20	55
8	30	58
6	40	60
10	33	70
9	25	66

a) Find the regression equation of X1 on X2 and X3, where X1 represents pull strength, X2 represents wire length and X3 represent the height.
b) Estimate X1 when X2 = 50 and X3 = 100.
c) Calculate standard error of estimate.
d) Calculate coefficient of determination and interpret the result.
e) Construct Anova table.
f) Test whether the model (Regression equation) is significant or not at 5% level of significance.
g) Test whether the independent variables X2 and X3 are significant.

➔Solution:

a)

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-7.862	4.169		-1.886	.156
	Wire length	-.049	.058	-.257	-.839	.463
	Height	.278	.082	1.041	3.396	.043

a. Dependent Variable: Pull Strength

We have; regression equation

X1 = b0+b1X2+b2X3

Substituting the value of b0, b1 and b2, we get;

X1 = -7.862-0.049*X1+0.277*X2

b)

When X2 = 50 and X3 = 100

X1 = -7.862-0.049*50+0.277*100

X1 = 17.338

Therefore, when X2 is 50 and X3 is 100 estimated value of X1 is 17.338

c)

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.907[a]	.823	.705	1.016

a. Predictors: (Constant), Height, Wire length

Therefore, the standard error of estimate is 1.016.

d)

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.907[a]	.823	.705	1.016

a. Predictors: (Constant), Height, Wire length

Coefficient of determination (R^2) = 0.823

=82.3%

The coefficient of determination is 82.3%. Hence, 82.3% of the total variation in variable X1(pull strength) is caused due to the variation in variable X2(wire length) and X3(height).

e) Required ANOVA Table:

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14.402	2	7.201	6.974	.074[b]
	Residual	3.098	3	1.033		
	Total	17.500	5			

a. Dependent Variable: Pull Strength

b. Predictors: (Constant), Height, Wire length

f) Null hypothesis: H0: β1 = β2 = 0 (The regression equation is not significant at 5% level of significance.)

Alternate hypothesis: H1: βi ≠ 0, i = 1, 2 (The regression equation is significant at 5% level of significance.)

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14.402	2	7.201	6.974	.074[b]
	Residual	3.098	3	1.033		
	Total	17.500	5			

a. Dependent Variable: Pull Strength

b. Predictors: (Constant), Height, Wire length

P value = 0.74

α = 5% = 0.05

Since, p value (0.74) > α (0.05). We accept H0.

Therefore, the regression equation is not significant at 5% level of significance.

g)

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-7.862	4.169		-1.886	.156
	Height	.278	.082	1.041	3.396	.043
	Wire length	-.049	.058	-.257	-.839	.463

a. Dependent Variable: Pull Strength

For Variable X2:

Null hypothesis:

H0: β1 = 0 (Variable X2 is not significant.)

Alternate hypothesis:

H1: β1 < 0 (Variable X2 is significant)

P value = 0.043

Let alpha (α) = 5% = 0.05

Since, p value (0.043) < α (0.05). We accept H1.

Therefore, the regression equation is significant at 5% level of significance.

For Variable X3:

Null hypothesis:

H0: β1 = 0 (Variable X3 is not significant.)

Alternate hypothesis:

H1: β1 < 0 (Variable X3 is significant)

P value = 0.463

Let alpha (α) = 5% = 0.05

➔Since, p value (0.463) > α (0.05). We accept H0.

Therefore, the regression equation is not significant at 5% level of significance.

Practical No 15: Date:- July/21/2023

The yield of treatments in different plots are as shown in the following plots. Carry out analysis.

T4	1401	T3	2536	T3	2459	T1	2537	T3	2827	T1	2069
T2	2211	T1	1797	T4	1170	T4	1516	T4	2104	T3	2385
T2	3366	T1	2104	T2	2591	T3	2460	T4	1077	T2	2544

➔Solution:

- ⇨ Null hypothesis: H0: μ1 = μ2 = μ3 = μ4 (There is no significant differences between the yield of treatments.)
- ⇨ Alternate hypothesis: H1: At least one μi is different where i = 1,2,3,4 (There is significant differences between the yield of treatments.)

ANOVA

value

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	4265689.961	3	1421896.654	11.253	.001
Within Groups	1768941.150	14	126352.939		
Total	6034631.111	17			

Let alpha (α) = 5% = 0.05

➔Since, p value (0.001) < α (0.05). We accept H1.

There is significant difference between the yields of treatments.